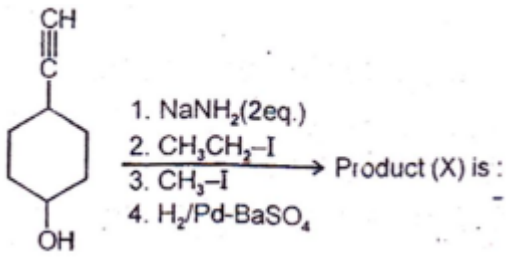
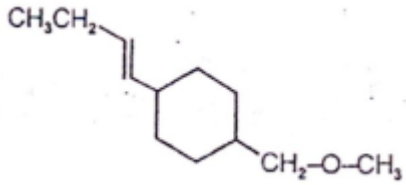
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

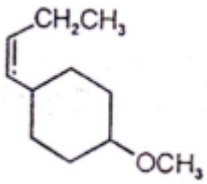
Q.1



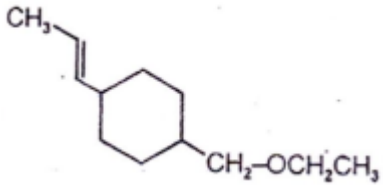
(A)



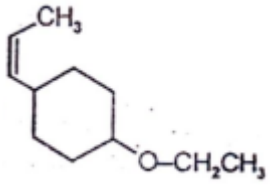
(B)



(C)

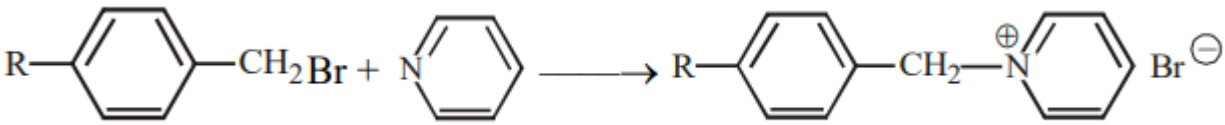


(D)



Q.2

The reaction of para substituted benzyl bromide with pyridine can be represented as



The reactivity order for different R (alkyl) groups in the above reaction is -

(A) $(CH_3)_3\,C- > (CH_3)_2\,CH- > CH_3-CH_2- > -CH_3$

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(B) $-CH_3 > CH_3-CH_2- > (CH_3)_2CH- > (CH_3)_3C-$

(C) $(CH_3)_2CH- > CH_3-CH_2- > -CH_3 > (CH_3)_3C-$

(D) $-CH_3 > (CH_3)_3C- > (CH_3)_2CH- > CH_3-CH_2-$

Q.3

The correct statement(s) about the following compound is/are : $CH_3-CHCl-CH=CH-CHCl-CH_3$

(A) The total number of stereoisomers possible is 6

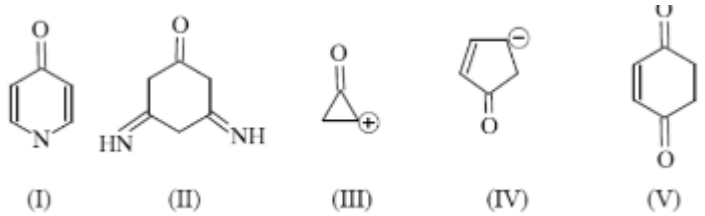
(B) The total number of diastereomers possible is 3

(C) The number of enantiomers possible for the trans isomer is 4

(D) The number of enantiomers possible for the cis isomer is 3

Q.4

In which of the following enolic form is more stable than keto form



(I) (II) (III) (IV) (V)

(A) I, II and III only

(B) I, III, IV and V only

(C) II, IV and V only

(D) I, II, III, IV and V

Q.5

What will be the total number of possible structural isomers with formula $C_7H_{12}Br_2$ fulfilling the following conditions:
(i) It should contain 6 membered cyclic ring.
(ii) None of isomer should contain bromine atom either on same or adjacent carbon atom.
(iii) None of the isomer should contain bromine atom on tertiary carbon atom

(A) 7

(B) 5

(C) 4

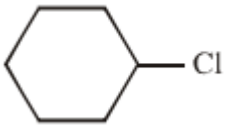
(D) 2

Q.6

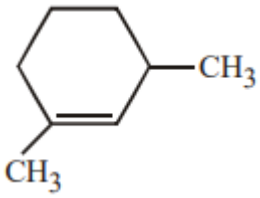
In which of the following molecules all the effects namely inductive, mesomeric and hyperconjugation operate -

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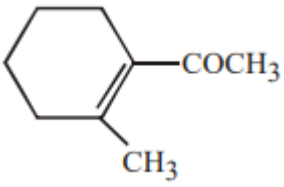
(A)



(B)



(C)



(D)



Q.7

Arrange the carbanions, $(CH_3)_3\bar{C}$, $\bar{C}Cl_3$, $(CH_3)_2\overline{C}H$, $C_6H_5\overline{C}H_2$ in order of their decreasing stability

(A) $C_6H_5\overline{C}H_2 > \overline{C}Cl_3 > (CH_3)_3\overline{C} > (CH_3)_2\overline{C}H$

(B) $(CH_3)_2\overline{C}H > \overline{C}Cl_3 > C_6H_5\overline{C}H_2 > (CH_3)_2\overline{C}$

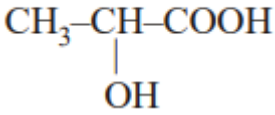
(C) $\overline{C}Cl_3 > C_6H_5\overline{C}H_2 > (CH_3)_2\overline{C}H > (CH_3)_3\overline{C}$

(D) $(CH_3)_2\overline{C} > (CH_3)_2\overline{C}H > \overline{C}H_2 > \overline{CCl_3}$

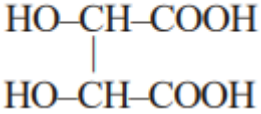
Q.8

Which of the following is not correctly matched : -

(A) Lactic acid



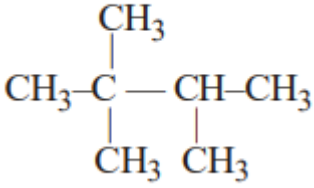
(B) Tartaric acid



(C) Pyvaldehyde

$CH_3C(CH_3)_2CHO$

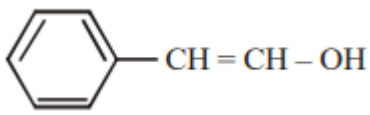
(D) Iso–octane

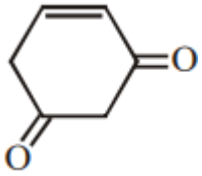


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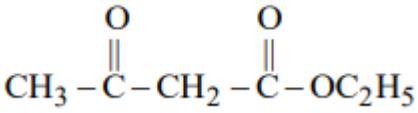
Q.9

Tautomerism is not exhibited by -

(A) 

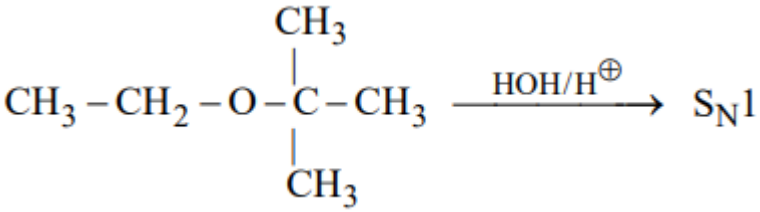
(B) 

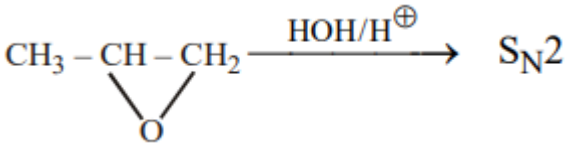
(C) 

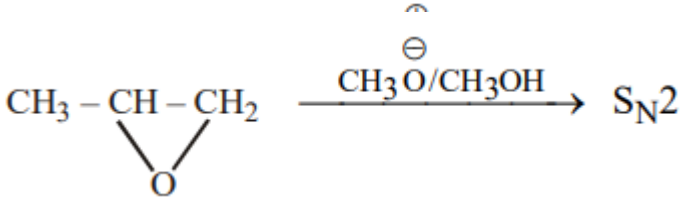
(D) 

Q.10

Choose the correct options

(A) 

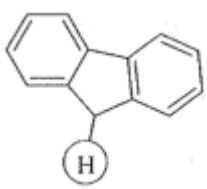
(B) 

(C) 

(D) 3 isomeric mono-substitution products theoretically are possible for the structure, $CH_2 = HC - CH_2 - CH_2 - CH = CH_2$

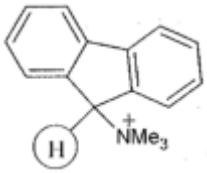
Q.11

Which of the following circled hydrogen is most acidic?

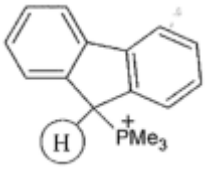
(A) 

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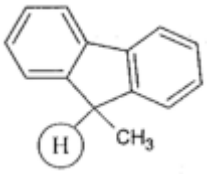
(B)



(C)



(D)



Q.12

Passage:

The compounds which have the same molecular and structural formulae but different arrangement of atoms or groups in space within their molecules are called stereo or space isomers. The phenomenon is known as space isomerism. The spatial arrangement of atoms or groups is also referred to as configuration of the molecule. Optical isomerism or entantiomerism is one of the stereoisomerism in which the molecules differ in rotation of plane polarised light. In order to exhibit optical activity the molecules should have asymmetric carbon and or molecular disymmetry.

Question: The number of configurational isomers possible for the compound $CH_3 - (CHCl)_2 - CH_3$ are -

(A) 8

(B) 10

(C) 14

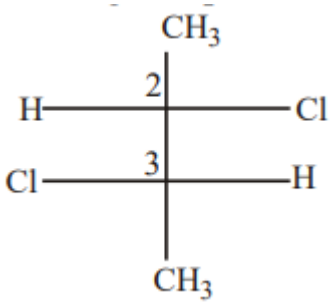
(D) 6

Q.13

Passage:

The compounds which have the same molecular and structural formulae but different arrangement of atoms or groups in space within their molecules are called stereo or space isomers. The phenomenon is known as space isomerism. The spatial arrangement of atoms or groups is also referred to as configuration of the molecule. Optical isomerism or entantiomerism is one of the stereoisomerism in which the molecules differ in rotation of plane polarised light. In order to exhibit optical activity the molecules should have asymmetric carbon and or molecular disymmetry.

Question: The absolute configuration of the following compound is



(A) $2\,S, 3R$

(B) $2\,S, 3\,S$

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(C) 2R, 3 S

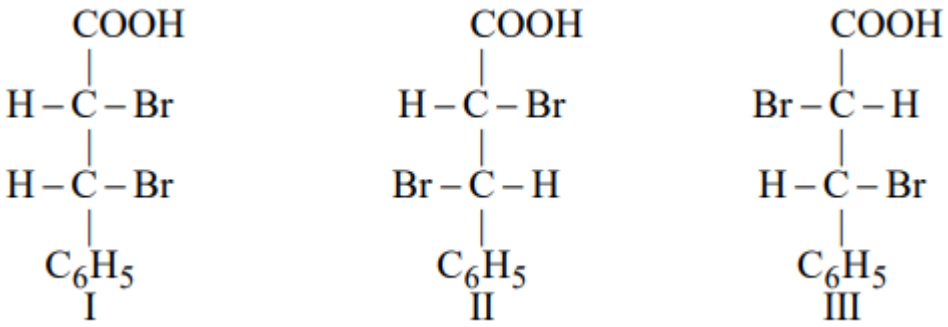
(D) 2R, 3R

Q.14

Passage:

The compounds which have the same molecular and structural formulae but different arrangement of atoms or groups in space within their molecules are called stereo or space isomers. The phenomenon is known as space isomerism. The spatial arrangement of atoms or groups is also referred to as configuration of the molecule. Optical isomerism or entantiomerism is one of the stereoisomerism in which the molecules differ in rotation of plane polarised light. In order to exhibit optical activity the molecules should have asymmetric carbon and or molecular disymmetry.

Question:



(A) I and II, I and III are diastereomeric pairs and II and III are enantiomers

(B) I and III, I and II, and II and III are diastereomers

(C) I and II, are enantiomers II and III are diastereomers

(D) II and III, and I and III are diastereomeric pairs

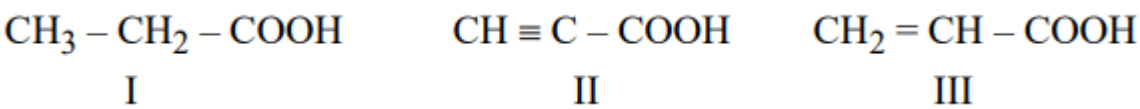
Q.15

Passage:

Substituents have definite effect on the strength of acids. Electron withdrawing substituents increase the strength of acid while electron donating substituents decrease the acid strength. In aromatic acids ortho substituents usually increase the strength regardless of its nature.

Hybridization of the carbon attached to the - COOH groups plays a part in the acidity of the acid. Chelation and consequent stabilization also plays an important part in increasing the acid strength.

Question: The strength of the acids I, II and III follow the order -



(A) I > III > II

(B) III > II > I

(C) II > III > *I*

(D) I > II > III

Q.16

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Passage:

Substituents have definite effect on the strength of acids. Electron withdrawing substituents increase the strength of acid while electron donating substituents decrease the acid strength. In aromatic acids ortho substituents usually increase the strength regardless of its nature.

Hybridization of the carbon attached to the - COOH groups plays a part in the acidity of the acid. Chelation and consequent stabilization also plays an important part in increasing the acid strength.

Question: The dicarboxylic acid which has the most acidic proton is -

(A) Succinic acid

(B) Tartaric acid

(C) Maleic acid

(D) Fumaric acid

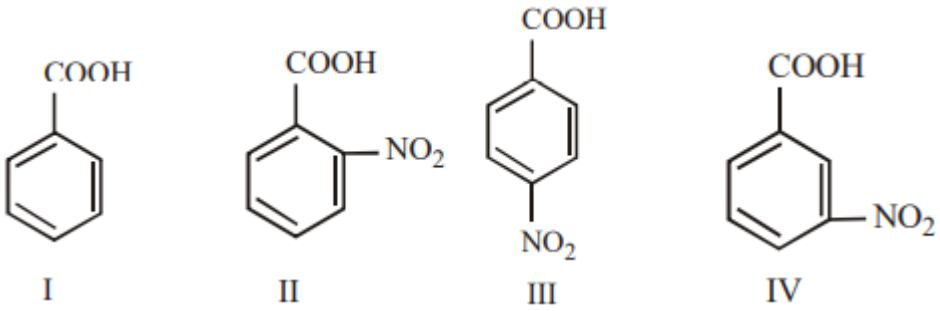
Q.17

Passage:

Substituents have definite effect on the strength of acids. Electron withdrawing substituents increase the strength of acid while electron donating substituents decrease the acid strength. In aromatic acids ortho substituents usually increase the strength regardless of its nature.

Hybridization of the carbon attached to the - COOH groups plays a part in the acidity of the acid. Chelation and consequent stabilization also plays an important part in increasing the acid strength.

Question: The correct order of decreasing strength of acid I, II, III and IV is -



I II III IV

(A) III > II > IV > I

(B) II > III > IV > I

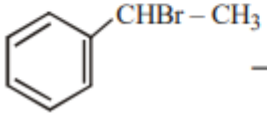
(C) II > IV > III > I

(D) III > IV > II > I

Q.18

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Match Column -

Column I	Column II
(A) $R_2CH-X \xrightarrow[C_2H_5OH]{C_2H_5OK}$	(p) S_N1
(B) $R_3C-X \xrightarrow[Heat]{C_2H_5OH}$	(q) S_N2
(C) $CH_3-CH_2-CHBr-CH_3 \xrightarrow[C_2H_5OH]{C_2H_5OK}$	(r) E_1
(D) 	(s) E_2

(A) A-(pqr) B-(pr) C-(pqrs) D-(r)

(B) A-(pqr) B-(pq) C-(pqr) D-(rp)

(C) A-(pqsr) B-(pr) C-(pqrs) D-(pr)

(D) A-(pqsr) B-(pr) C-(pqrs) D-(p)

Q.19

The number of optically active isomers of the following compound is

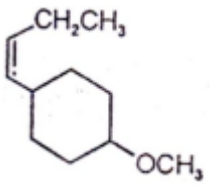
$OHC-CHBr-CHOH-CHBr-CH_2OH$

Q.20

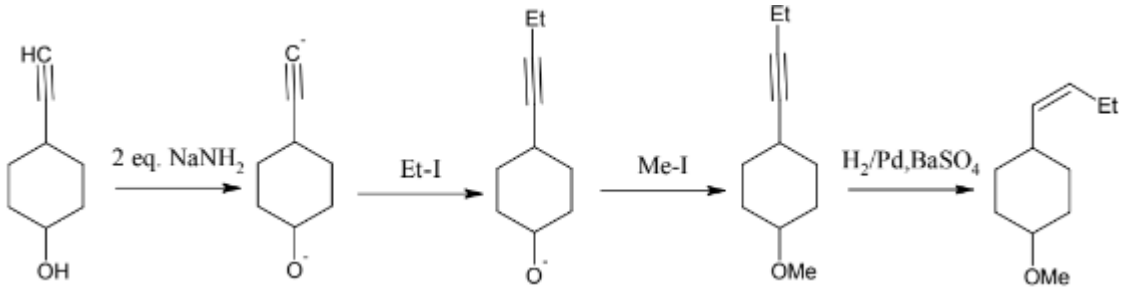
A compound containing C, H and O with molecular weight 122 on acetylation gives the derivative whose molecular weight is 290 . The no. of - OH groups present in the original molecule is

Answers & Solutions

Q.1 Answer:



Solution:



Q.2 Answer:

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$-CH_3 > CH_3 - CH_2- > (CH_3)_2 CH- > (CH_3)_3 C-$

Solution:

The electron releasing tendency of alkyl group when attached to benzene and unsaturated sytem is Methy $>$ ethyl $>$ isopropyl $>$ t-butyl. The electron release in such cases is due to hyperconjugation. It is the study of the rates of this reaction that led Baker and Nathan to puprose the concept of hyperconjugation.

Q.3 Answer:

The total number of stereoisomers possible is 6

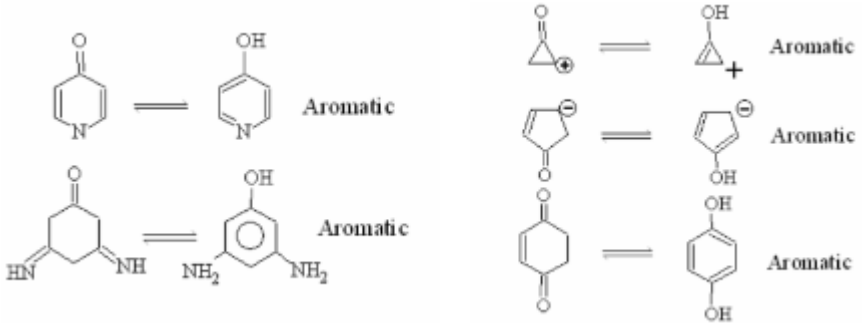
Solution:

The compound can have two geometrical isomers (cis and trans). Since the compound has two similar chiral carbon atoms, each geometrical isomer can have three isomers. So, 6 stereoisomers are possible. They are R cis R, S cis S, R cis S, R trans R, S trans S and R trans S.
For the cis isomer, R cis R and S cis S are enantiomers, i.e. 2 enantiomers are possible. Only two enantiomers are possible for the trans isomer. So statement (C) is wrong. The total number of diastereomers possible is 12. So statement (B) is also wrong.

Q.4 Answer:

I, II, III, IV and V

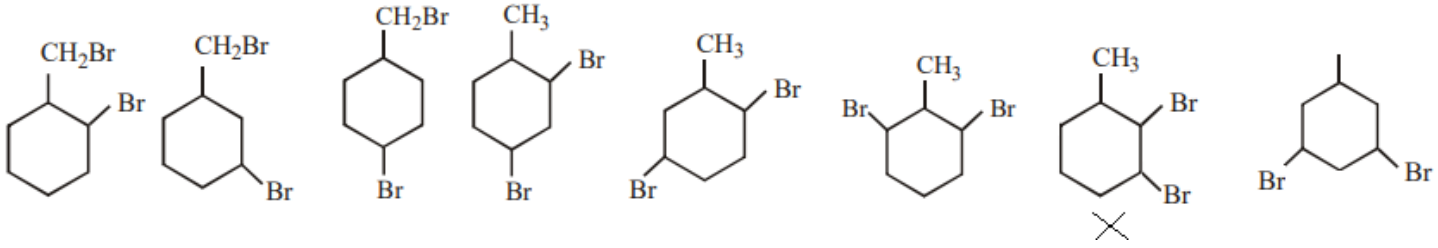
Solution:



Q.5 Answer:

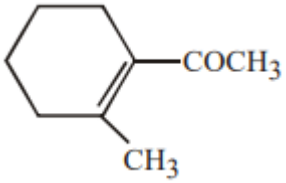
7

Solution:

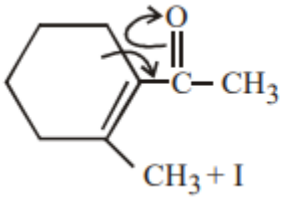


Q.6 Answer:

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Solution:



has all effect inductive, mesomeric and hyperconjugation.

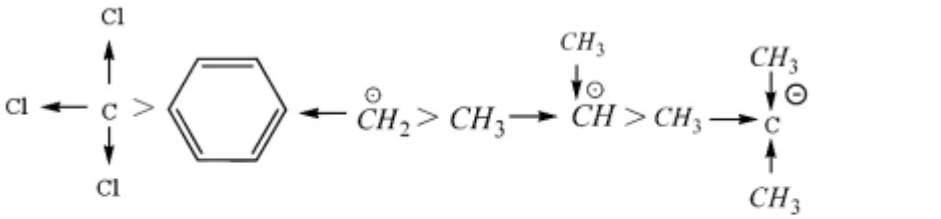
Q.7 Answer:

$\overline{C}Cl_3 > C_6H_5\overline{C}H_2 > (CH_3)_2\,\overline{C}H > (CH_3)_3\,\overline{C}$

Solution:

1 effect $[e^-$ withdrawing] exerting groups stabilizes carbanion by the dispersal of their negative charge, while $+I$ effect exerting $[e^-$ releasing] groups destabilizes the carbanion by increasing electron density on them.

On the other hand, resonance stabilized carbanion are stable due to the involvement of their lone pair of electron with the delocalization of π-electrons of attached phenyl group. Thus,



(3- I effect exerting group of C l)

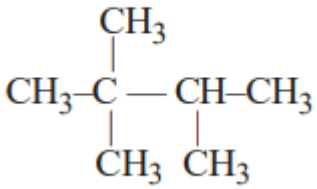
(-I effect of phenyl group as well as delocalisation of electrons)

(+1 effect of 2- CH_3 group)

(+1 effect of 3- CH_3 groups)

Q.8 Answer:

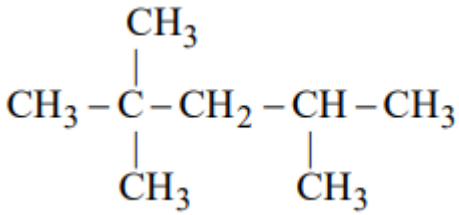
Iso–octane



Solution:

.

Iso-octane is



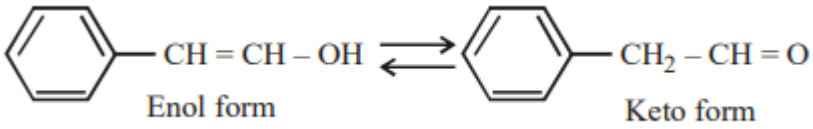
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Q.9 Answer:

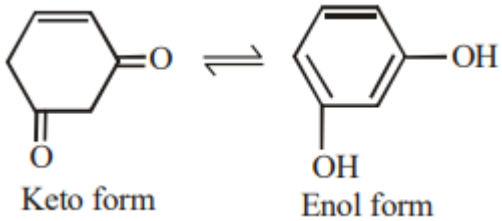


Solution:

(A)

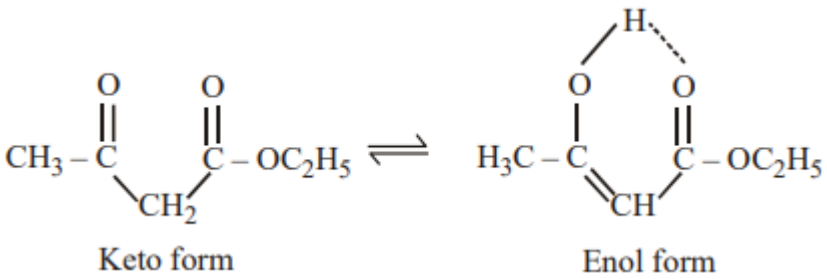


(B) (Tautomerism is exhibited because of stabilization by aromatisation)

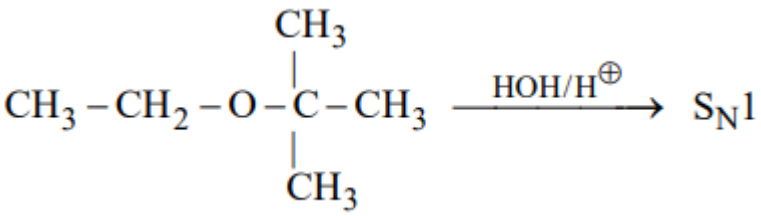


(C) Does not have hydrogen at a saturated α carbon.

(D)



Q.10 Answer:

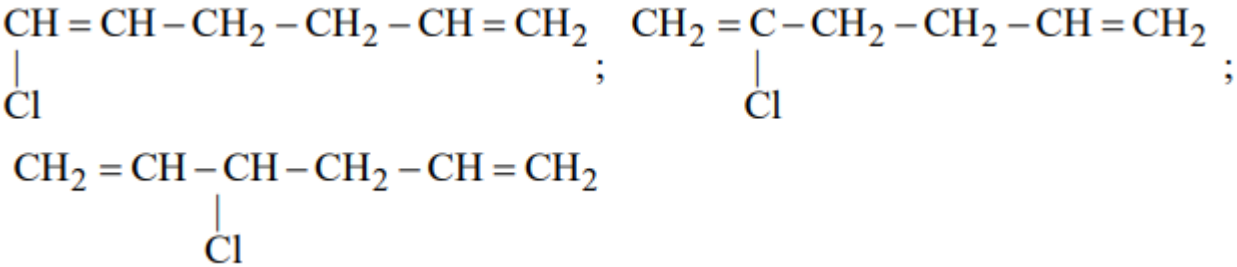


Solution:

(A) and (B) acidic medium favour S_N1 mechanism.

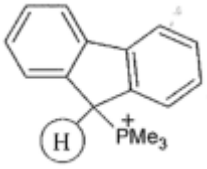
(C) $CH_3O^{\ominus}$ is base and basic medium favours S_N2 mechanism.

(D)

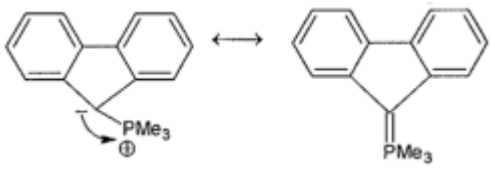


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Q.11 Answer:



Solution:



Negative charge can be delocalised on phosphorous also in addition to phenyl rings. So the anion is more stabilised by resonance.

Q.12 Answer:

10

Solution:

(i) When compound has even number of carbon atoms $n = 4$

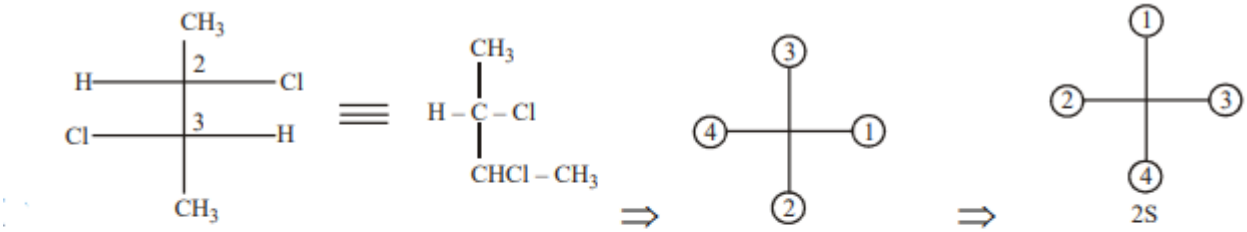
No. of optically active forms $= a = 2^{n-1}$

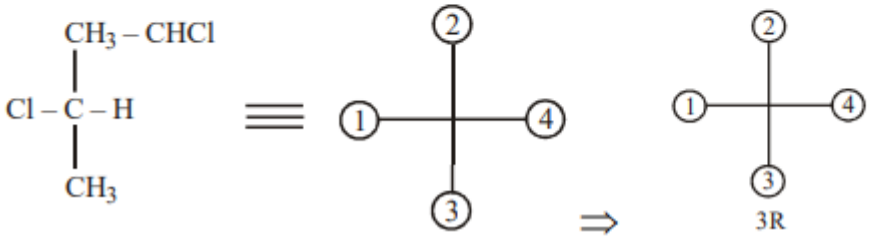
$a = 2^{4-1} = 2^3 = 8$

Number of meso form, $m = 2^{(\frac{n}{2}-1)} = 2^{\frac{4}{2}-1} = 2^{2-1} = 2$

Total number of configurational isomers $(= a + m = 8 + 2 = 10$

(ii)





(iii) II and III are optically active mirror images - enantiomers I and III, I and II are diasteromeric pairs.

Q.13 Answer:

2 S, 3 S

Solution:

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(i) When compound has even number of carbon atoms $n = 4$

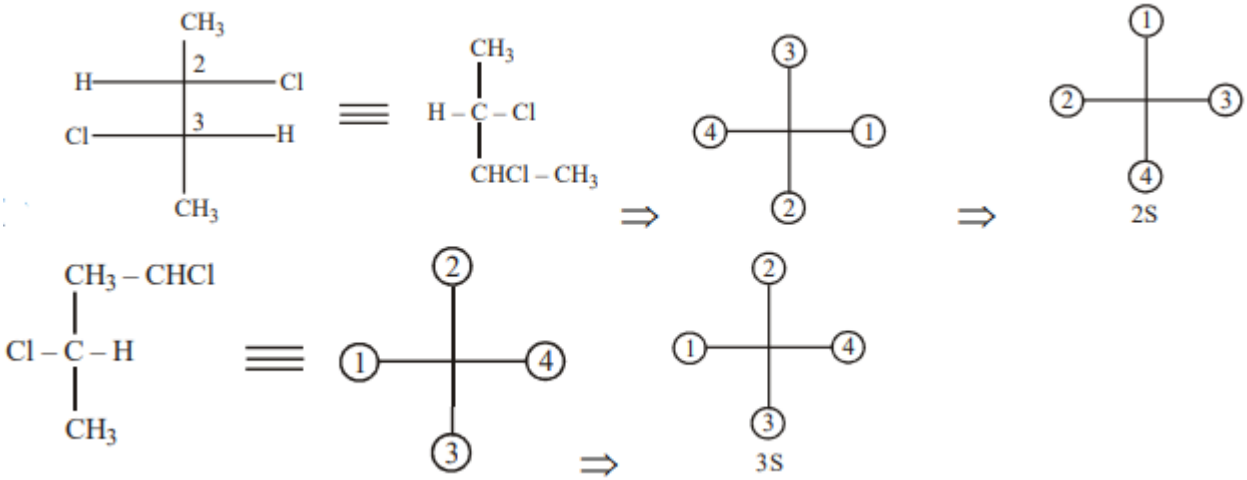
No. of optically active forms $= a = 2^{n-1}$

$a = 2^{4-1} = 2^3 = 8$

Number of meso form, $m = 2(n/2 - 1) = 2^{4/2} - 1 = 2^{2-1} = 2$

Total number of configurational isomers $(= a + m = 8 + 2 = 10$

(ii)



(iii) II and III are optically active mirror images - enantiomers I and III, I and II are diasteromeric pairs.

Q.14 Answer:

I and II, I and III are diastereomeric pairs and II and III are enantiomers

Solution:

(i) When compound has even number of carbon atoms $n = 4$

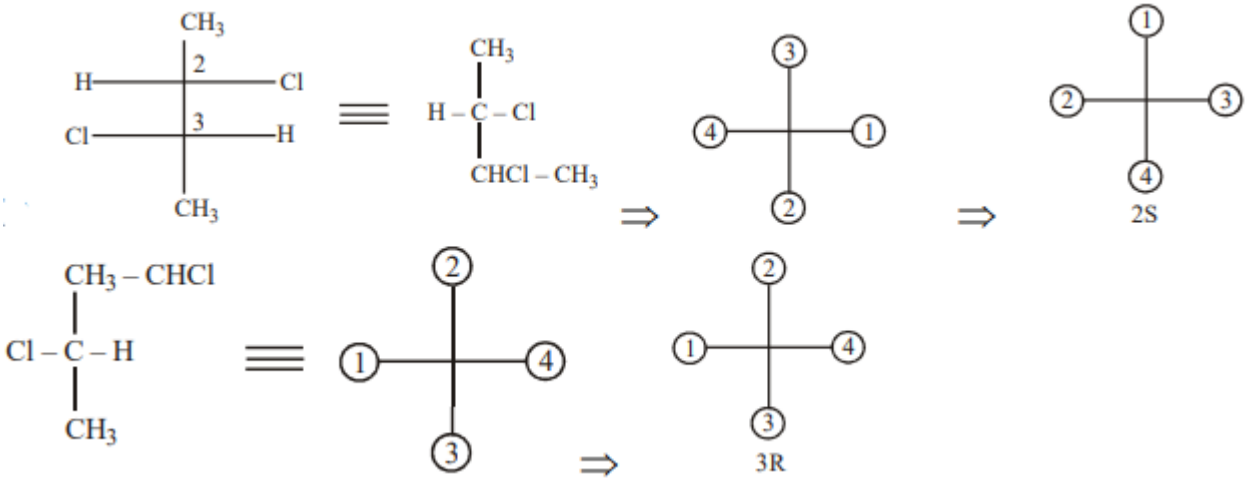
No. of optically active forms $= a = 2^{n-1}$

$a = 2^{4-1} = 2^3 = 8$

Number of meso form, $m = 2(n/2 - 1) = 2^{4/2} - 1 = 2^{2-1} = 2$

Total number of configurational isomers $(= a + m = 8 + 2 = 10$

(ii)



(iii) II and III are optically active mirror images - enantiomers I and III, I and II are diasteromeric pairs.

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Q.15 Answer:

$\text{II} > \text{III} > I$

Solution:

(i)

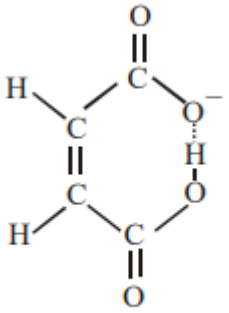
$$\underset{\text{sp hybridized carbon}}{\underset{\downarrow}{CH \equiv C - COOH}} > \underset{\text{sp}^2 \text{ hybridized carbon}}{\underset{\downarrow}{CH_2 = CH - COOH}} > \underset{\text{sp}^3 \text{ hybridized carbon}}{\underset{\downarrow}{CH_3 - CH_2 - COOH}}$$

Greater the 's' character, greater will be the strength of acid. sp hybridized carbon has 50% 's' character. sp^2 hybridized carbon has 33.3% and sp^3 hybridized carbon has 25% s character.

sp hybridized carbon is more electronegative than sp^2 and sp^3 carbon.

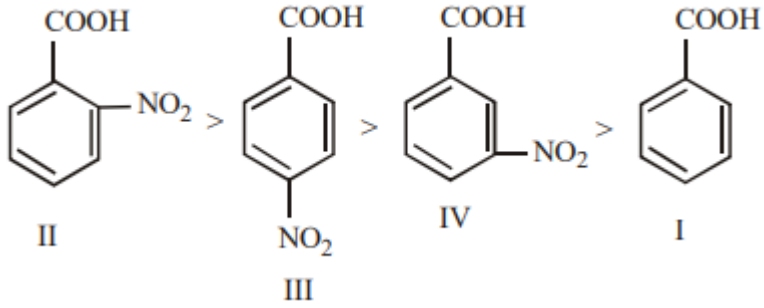
(ii) Maleic and fumaric acids have sp^2 hybridized carbon and hence are stronger than succinic and tartaric acid.

Among the two, maleic acid (cis) is stronger than fumaric acid (trans) because the monoanion is stabilized by chelation which is not possible in fumaric acid (trans)



Maleic acid (cis)

(iii) stabilization due to chelation by hydrogen bonding.



In aromatic acid whether it is $+\text{I}$ or $-\text{I}$ group, ortho substituted acid are stronger than benzoic acid. Since the ortho substituent is NO_2 group, both - I and $-\text{M}$ effects will cause increase in the strength of acids.

It is followed by the acid with a para nitro group, meta nitro group and finally the unsubstituted acid.

Q.16 Answer:

Maleic acid

Solution:

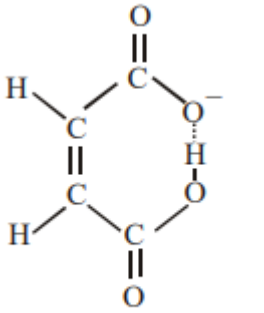
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(i)

$CH \equiv C - COOH$ (sp hybridized carbon) $>$ $CH_2 = CH - COOH$ (sp^2 hybridized carbon) $>$ $CH_3 - CH_2 - COOH$ (sp^3 hybridized carbon)

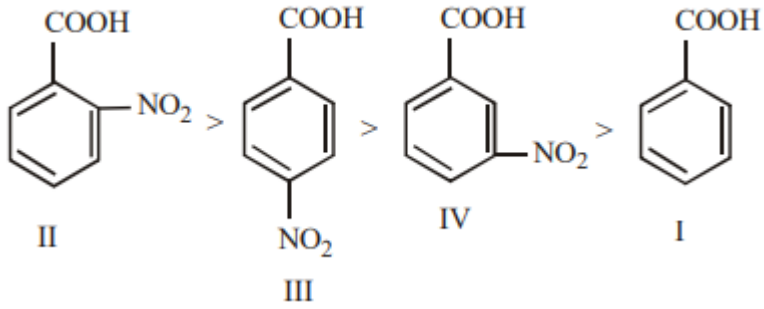
Greater the 's' character, greater will be the strength of acid. sp hybridized carbon has 50% 's' character. sp^2 hybridized carbon has 33.3% and sp^3 hybridized carbon has 25% s character. sp hybridized carbon is more electronegative than sp^2 and sp^3 carbon.

(ii) Maleic and fumaric acids have sp^2 hybridized carbon and hence are stronger than succinic and tartaric acid. Among the two, maleic acid (cis) is stronger than fumaric acid (trans) because the monoanion is stabilized by chelation which is not possible in fumaric acid (trans).



Maleic acid (cis)

(iii) stabilization due to chelation by hydrogen bonding.



In aromatic acid whether it is $+I$ or $-I$ group, ortho substituted acid are stronger than benzoic acid. Since the ortho substituent is NO_2 group, both - I and $-M$ effects will cause increase in the strength of acids. It is followed by the acid with a para nitro group, meta nitro group and finally the unsubstituted acid.

Q.17 Answer:

II > III > IV > I

Solution:

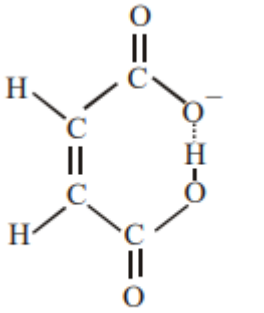
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(i)

$CH \equiv C - COOH$ (sp hybridized carbon) $>$ $CH_2 = CH - COOH$ (sp^2 hybridized carbon) $>$ $CH_3 - CH_2 - COOH$ (sp^3 hybridized carbon)

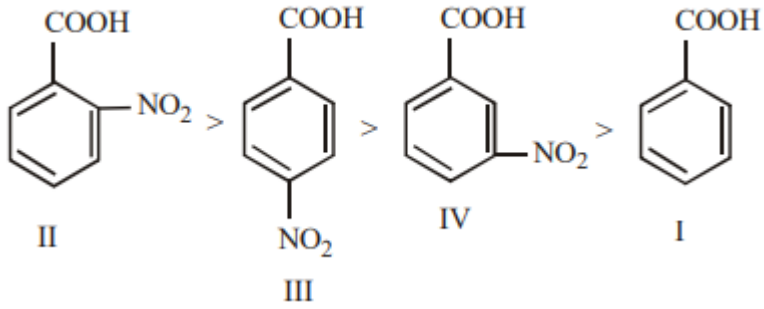
Greater the 's' character, greater will be the strength of acid. sp hybridized carbon has 50% 's' character. sp^2 hybridized carbon has 33.3% and sp^3 hybridized carbon has 25% s character. sp hybridized carbon is more electronegative than sp^2 and sp^3 carbon.

(ii) Maleic and fumaric acids have sp^2 hybridized carbon and hence are stronger than succinic and tartaric acid. Among the two, maleic acid (cis) is stronger than fumaric acid (trans) because the monoanion is stabilized by chelation which is not possible in fumaric acid (trans).



Maleic acid (cis)

(iii) stabilization due to chelation by hydrogen bonding.



In aromatic acid whether it is $+I$ or $-I$ group, ortho substituted acid are stronger than benzoic acid. Since the ortho substituent is NO_2 group, both - I and $-M$ effects will cause increase in the strength of acids. It is followed by the acid with a para nitro group, meta nitro group and finally the unsubstituted acid.

Q.18 Answer:

A-(pqsr) B-(pr) C-(pqrs) D-(pr)

Solution:

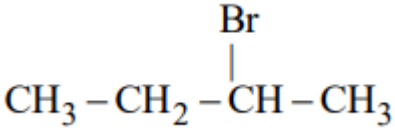
$(A) - pqrs, (B) - pr, (C) - pqrs, (D) - pr.$

(A) $R_2CH - X$ undergoes S_N1, E_1, S_N2 and E_2 in presence of C_2H_5OK and C_2H_5OH. But there is no inversion of configuration.

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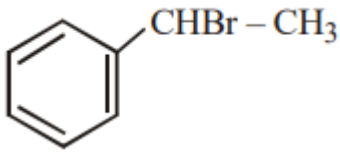
(B) $R_3C - X$ **undergoes solvolysis by** S_N1 **and also** E_1**.**

(C)



(being a 2° **halide) undergoes** $S_N1, E_1,\ S_N2$ **and** E_2 **reactions with** $C_2H_5O^-$ **and** C_2H_5OH **(solvent).** S_N2 **reaction leads to inversion of configuration.**

(D)

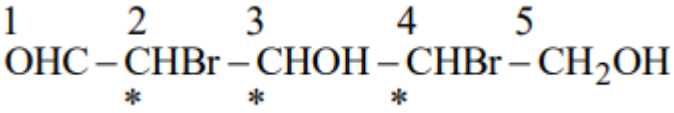


undergoes S_N1 **and** E_1 **with polar solvent (**$HCOOH$ **and water).**

Q.19 Answer:

—

Solution:



C_2, C_3, C_4 **are asymmetric carbons**

$n = 3$ **odd and** $a = 2^n = 2^3 = 8$**, the molecule cannot be divided into two identical halves.**

Q.20 Answer:

—

Solution:

Molecular weight of compound $= 122\text{amu}$

Molecular weight of acetyl derivative $= 290$ **amu**

$\therefore$ **Increase in mass due to acetylation** $= 290 - 122 = 168\text{amu}$

Increase in mass due to acetylation of one $-OH$ **group** $= 42\text{amu}\ (CH_3CO - H)$

Hence, no. of $-OH$ **groups present in original molecule** $= \frac{168}{42} = 4$

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